THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 160 cm	320 - 160.
2	**(c)** P	The pin falls first.
3	**(d)** 50 °C	Linear scale with equal spacing gives 50 °C.
4	**(b)** heat is used to change the state	Latent heat idea.
5	**(a)** time period is 1.8 s in both	T = 36/20 = 1.8 and 18/10 = 1.8.
6	**(c)** B	Translucent materials.
7	**(a)** 40 s	Twice slower.
8	**(a)** 100 g	Mass is conserved.
9	**(a)** 10.5 cm	12.5 - 2.
10	**(a)** its shape displaces much water	Hollow shape lowers average density.
11	**(d)** aluminium and plastic	Pan conducts, handle insulates.
12	**(d)** it changed to vapour and mixed with air	Matter is conserved.
13	**(b)** 850 m	2700 - 1850.
14	**(d)** S	The pin falls last.
15	**(c)** metal cup	Metal conducts heat away fastest.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	3.6 km = 360000 cm (1); 360000 / 60 = 6000 steps (1).	1+1
17	Empty bottle has low average density (1); Sand makes it heavier (1).	1+1
18	Temperatures are equal (1); The larger quantity contains more heat energy (1).	1+1
19	Vapour cannot escape (1); Air in the bag becomes saturated (1).	1+1
20	0.003 km = 3 m (1); 3 m = 300 cm (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	1.1 km = 1100 m (1); 1100 / 2.2 (1); 500 turns (1).	1+1+1
22	It is a metal (1); Shiny (lustrous) and sinks (1); Conducts heat along its length (1).	1+1+1
23	Atmospheric pressure is lower on hills (1); Water boils at a lower temperature (1); Therefore reading differs (1).	1+1+1
24	4 x 30 = 120 L (1); 120/2000 (1); 6 per cent (1).	1+1+1
	OR: Use a flexible tape (1); Follow the curve (1); Verify with thread (1).	1+1+1
25	Prevent shock (1); Rubber or plastic (1); Insulate the hand (1).	1+1+1
	OR: Reduces loss of hot air (1); Stops steam escaping (1); Keeps heat (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Tree trunk: measuring tape or thread around (1); Then measure thread (1); Coin: stack 20 coins and measure (1); Divide by 20 (1); Pencil: ruler with zero at one end (1).	1+1+1+1+1
	OR: Same heat source and time (1); Measure temperature of the free end (1); Lowest rise is best (1); Use tongs and avoid burns (1); Rank and justify (1).	1+1+1+1+1
27	Same volume, same starting temperature (1); Same containers (1); Record temperature every 2 min (1); Plot a graph or table (1); Compare cooling (1).	1+1+1+1+1

	OR: Identical wet cloths (1); Place in three locations (1); Weigh at intervals (1); Greatest loss shows the fastest drying (1); Conclusion with reasons (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** melting and boiling	States change.
2	**(c)** it is in linear motion and its speed increases	It falls along a straight line, speeding up.
3	**(a)** C conducts heat and D does not	Test for conduction.
4	**(b)** the 20 °C object	Heat enters the colder one.
5	**(d)** 10 mL	50/5.
6	**(a)** 8.3 cm	All readings are higher.
7	**(c)** it is light and cushions shocks	Soft and light.
8	**(d)** metal rod hotter	Metal conducts quickly.
9	**(b)** smaller volume and large surface to depth ratio	Less water.
10	**(d)** 1200 m	2.5 x 480.
11	**(b)** iron is heavier for the same size	Comparing same volume.
12	**(c)** heat in equals heat out	Balance.
13	**(c)** the colour of the container	Not relevant.
14	**(a)** a rolling ball	Rolling ball.
15	**(c)** using a magnet	Magnet only affects iron; keeps others intact.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Rubber is an electrical insulator (1); Prevents electric shock (1).	1+1
17	Alcohol does not freeze easily (1); It is safer (1).	1+1
18	Air's vapour touches the cold surface (1); It condenses on the outside (1).	1+1
19	Swing repeats at equal intervals (1); Leaf does not repeat (1).	1+1
20	Shaping changes the object (1); E.g. steel gives spoon, pan or bridge parts (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Moist air rises over hills (1); It cools and condenses (1); Clouds drop rain (1).	1+1+1
22	40 / 5 (1); 8 cm (1); Scale used (1).	1+1+1
23	Metals are strong (1); Metals give a ringing sound when struck (1); Metals can be shaped (1).	1+1+1
24	Touching the bottom (1); Reading from an angle (1); Not waiting for a steady reading (1).	1+1+1
	OR: Trees release water vapour into the air (1); Roots hold the soil and help water to seep into the ground (1); Fewer floods and more groundwater (1).	1+1+1
25	Coil length 8 cm = 80 mm (1); 80 / 25 (1); 3.2 mm (1).	1+1+1
	OR: Wrap one hot container in cotton and other bare (1); Measure temperature after 15 min (1); Cotton traps air and keeps hot longer (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	One reading may have error (1); Repeat several times (1); Take average (1); Avoid parallax (1); Use a good scale (1).	1+1+1+1+1
	OR: Take a steel foil (1); Place flat: sinks (1); Shape as a boat (1); It floats (1); Because shape holds air (1).	1+1+1+1+1

27	Matka: evaporation (1); Quilt: trapped air (1); Mud house: poor conductor (1); Fridge: removes heat (1); Thermos: insulates (1).	1+1+1+1+1
	OR: Ice warms to 0 °C (1); Flat at 0 °C: melting (1); Water warms to 100 °C (1); Flat at 100 °C: boiling (1); Heat used for change of state (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** normal temperature with small variation	Normal range.
2	**(b)** 60 L	2 x 30.
3	**(a)** to reduce random errors	Average.
4	**(a)** it may rust or conduct heat	Metals conduct heat and may rust.
5	**(b)** electric iron handle	Handle plastic.
6	**(c)** melting of ice	Water can be refrozen.
7	**(a)** 47 cm	188/4.
8	**(a)** sugar and chalk powder both look white but behave differently in water	One dissolves, the other does not.
9	**(b)** the water is at the same temperature as the room	Same as surroundings.
10	**(a)** increase	Larger area.
11	**(d)** each blade moves in a circular path	Blades go round.
12	**(b)** sand on the filter paper	Sand is insoluble.
13	**(a)** 20 °C	Below its range.
14	**(a)** 180 mL	3600/20.
15	**(c)** placed touching the object and viewed straight	Reduce gap.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	The bucket can have more heat (1); Temperature shows hotness, not amount (1).	1+1
17	Fall (1); Wells may dry (1).	1+1
18	The edge is curved (1); Use a thread or tape and then measure it (1).	1+1
19	Metal conducts heat (1); Plastic is an insulator (1).	1+1
20	No, both are at room temperature (1); Metal conducts heat from the hand faster (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Same amount of water at same temperature (1); Wrap each beaker (1); Measure after 30 minutes (1).	1+1+1
22	Cover a bowl and an open bowl (1); Open bowl loses level (1); Covered shows droplets on the lid (1).	1+1+1
23	Rotation: spin about own axis (1); Top, fan (1); Revolution: motion around another point; Earth around Sun, Moon around Earth (1).	1+1+1
24	Light (1); Cheap (1); Waterproof and can be moulded (1).	1+1+1
	OR: Record patterns (1); Notice rise or fall (1); Use trends for prediction (1).	1+1+1
25	Temperature is below 0 °C (1); Snow is compact (1); Insulates inside (1).	1+1+1
	OR: Different reference (1); Passenger moves with the train (1); Observer is fixed to the ground (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	T1 = 60/30 = 2 s (1); T2 = 30/30 = 1 s (1); Shorter pendulum has smaller period (1); Length ratio 4:1, time ratio 2:1 (1); Conclusion: time period depends on length (1).	1+1+1+1+1
	OR: Magnet for iron (1); Add water to dissolve salt and dye (1); Filter for sand (1); Evaporate to get salt and dye mix (1); Principle: magnetism, solubility, evaporation (1).	1+1+1+1+1

27	Particles at the hot end gain energy (1); Pass on to neighbours (1); Heat travels to the cooler end (1); Plastic is an insulator (1); Slow transfer (1).	1+1+1+1+1
	OR: Sea water in a basin (1); Sun heats, evaporation (1); Vapour condenses on the cool cover (1); Drops collect in a container (1); Salt left behind (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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